Multi-modality Heterogeneous Web Source Integration

by E. Chang, UCSB; J. Cho, UCLA; C. Li, UC Irvine

In this project, we intend to develop novel technologies that enable users to access high-quality, timely information from the Web in an intuitive and comprehensive fashion. To achieve this goal, our project will build an integrated system consisting of three main research thrusts: 1) *effective data collection*, 2) *multi-modality data representation*, and 3) *intelligent query-concept formulation*. The PIs of this project are three computer scientists from the University of California with complementary and synergistic backgrounds in information technologies. Our research will tackle the following three obstacles that make quality-information access difficult for users: 1) the growing number of stale and recondite pages not presently indexed by search systems; 2) the limitations of keyword-only representation employed by search systems at present; and 3) the difficulties experienced by most users in articulating queries for searches.

In our first research thrust, *effectve data collection*, we will study mechanisms for tracking changes on the Web and refreshing our data frequently and efficiently. Our system will be much more sophisticated than the monitoring systems that track only bookmarked sites. We will explore ways to integrate dynamic Web information, so that users can access more information seamlessly through a uniform interface. We will also develop algorithms for discovering information that is most relevant to a query, and for retrieving such information intelligently. In our second research thrust, *multi-modality data representation*, we plan to leverage multimedia information on the Web to enrich page representation. To achieve this objective, we will first develop a multi-modality model. When a search is conducted, meta-data in both text and multimedia content will be leveraged to improve search results. In particular, we propose three solutions: using image-category labels to enrich text keywords, mining surrounding text to annotate images, and leveraging meta-data of both modalities to improve the search results. In our third thrust, *intelligent query-concept formulation*, we plan to design novel active-learning algorithms which will make it easy for users to formulate queries. We plan to extend our active-learning algorithms, as well as to develop new algorithms for multi-modality query-concept learning.

We believe that this project will have a substantial impact on helping users access high-quality and timely Web information. Our intelligent query-concept learners will make it more intuitive to describe users' information needs and to capture their query concepts more precisely. In addition, by incorporating multimedia information for data representation, we will enable users to interact with the system using pictures and graphics, as well as Boolean keyword expressions. As a result, even young students will be able access information from the Web with ease. Librarians and teachers will have access to more information, more easily obtained. Inventors can do their own patent searches inexpensively. The fruit of this project will have many social benefits, not the least of which is to reduce "information inequality" by making important data available to all, including underprivileged individuals.